DOCUMENTATION

Project Title: Onsite Event Registration System (School - Based)

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System Documentation
BSCS - 1A

I. Application Features (CO1, CO2)

The Onsite Event Registration System provides a complete CRUD (Create, Read, Update, Delete) suite.

Add (Create): Register walk-in students; automatically generates ticket numbers and timestamps.

View (Read): Displays data in responsive JTable components with real-time capacity statistics.

Edit (Update): Allows modification of event details (name, date, max attendees) and attendee records.

Delete/Cancel: Provides a "Cancel" status to free up seating capacity without losing historical data.

Search & Filter: Real-time filtering via DocumentListener and status-based dropdowns.

Forms (JFrames): Includes LoginFrame, HomeFrame, EventDashboardFrame, and multiple JDialogs for registration and editing.

II. Data Persistence (CO2)

The system ensures all data is saved locally without requiring an internet connection.

JSON Integration: Uses the json-simple-1.1.1.jar library.

Storage Logic: The JsonHandler class converts Java ArrayLists into JSON arrays, persisting data to the data/ directory.

Error Handling: Implements try-catch blocks to handle IOException or ParseException during file access.

III. Error Handling and Code Quality (CO1, CO2)

Authentication: The system checks credentials and provides visual cues if a login fails.

Data Integrity: We enforce strict rules, such as requiring the YYYY-MM-DD date format and preventing empty forms.

Safety Checks: The system includes a Capacity Limit feature that blocks new registrations once an event is full, and Duplicate Detection to ensure the same person doesn't sign up twice for the same event.

IV. Documentation (CO3)

A. Introduction

What is the System?

The School-Based Onsite Event Registration System is a Java Swing desktop application designed specifically to handle walk-in registrations for school activities such as departmental seminars, assemblies, and intramural events. It provides a 100% offline, zero-latency solution to manage event attendance.

Purpose of the Application

The primary purpose of this application is to streamline onsite school event registration. By transitioning from paper-based lists to a digital system, it efficiently manages participant data, significantly reduces manual administrative work, and improves data accuracy. Furthermore, it makes tracking the attendance of registered students faster and easier, while automatically enforcing seating capacities and preventing duplicate registrations.

Target Users

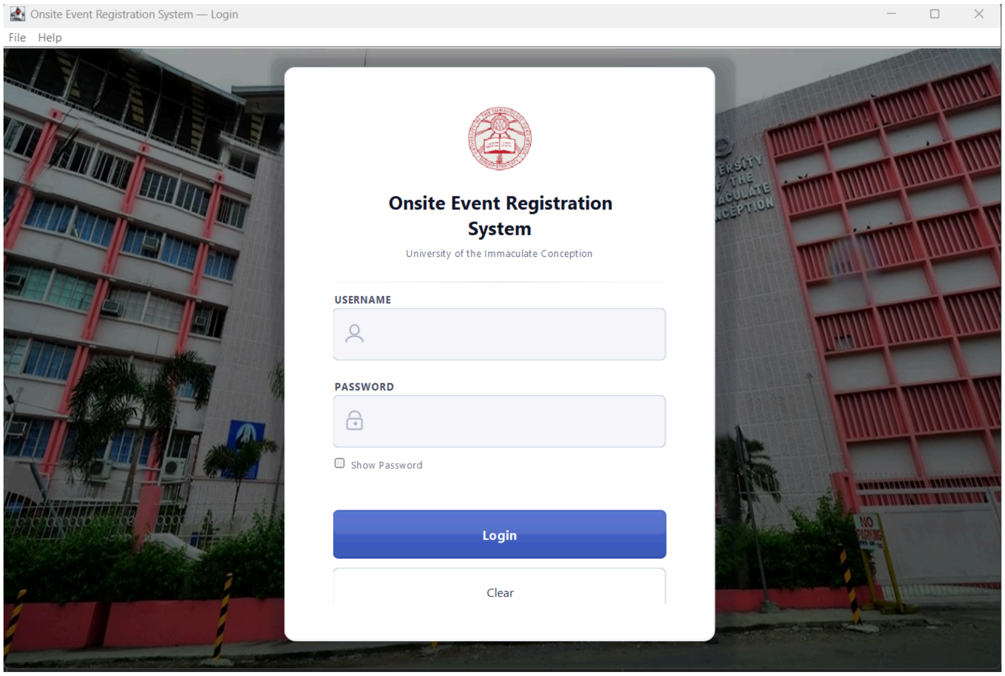
This system is designed to be used by the **System Administrator** (e.g., Student Council Officers) or a **Selected Operator** assigned to man the registration desk during the day of the event.

B. User Manual

This section provides a step-by-step guide on how to navigate and operate the system.

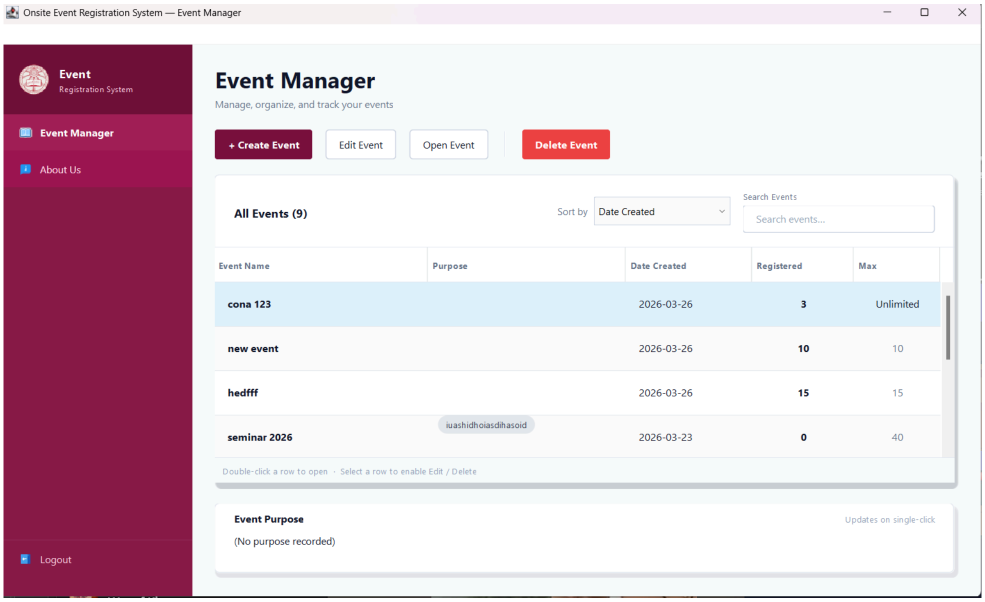
1. System Login

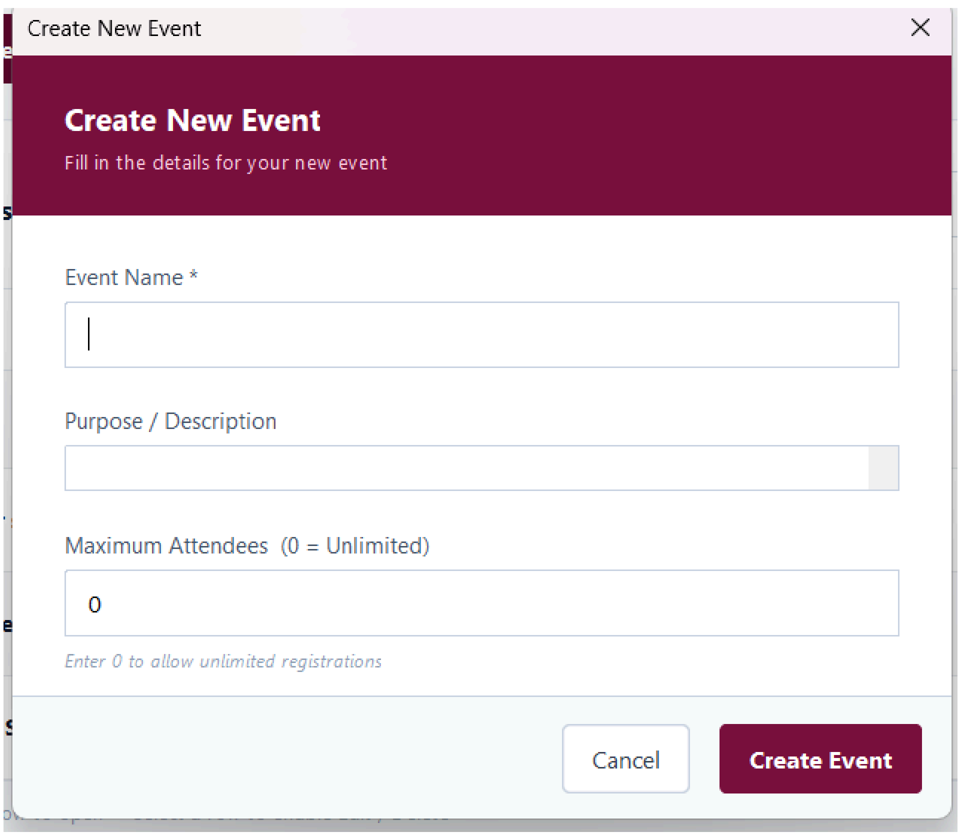
1. Launch the application. The LoginFrame will appear.
2. Enter the authorized Username and Password using (user: admin, pass: admin123).
3. Click the **"Login"** button to authenticate and access the system.



2. The Home Dashboard (Managing Events)

1. Upon successful login, you are redirected to the HomeFrame, which serves as the main dashboard showing all events.
2. **To Create an Event:** Click the **"+ New Event"** button. A dialog will appear asking for the Event Name, Purpose, Date, and Maximum Attendees. Fill this out and click "Save".
3. **To Search an Event:** Use the "Search events..." bar at the top to quickly find an event by name or purpose.
4. **To Open an Event:** Locate your event in the table and click the **Action button (Open Dashboard) or double click** to start registering students for that specific event.


Onsite Event Registration System — Event Manager
Event
Registration System
Event Manager
About Us
Logout
Event Manager
Manage, organize, and track your events
+ Create Event
Edit Event
Open Event
Delete Event
All Events (9)
Sort by
Date Created
Search Events
Search events...
Event Name
Purpose
Date Created
Registered
Max
cona 123
2026-03-26
3
Unlimited
new event
2026-03-26
10
10
hedfff
2026-03-26
15
15
seminar 2026
iuashidhoiasdihasoid
2026-03-23
0
40
Double-click a row to open · Select a row to enable Edit / Delete
Event Purpose
Updates on single-click
(No purpose recorded)

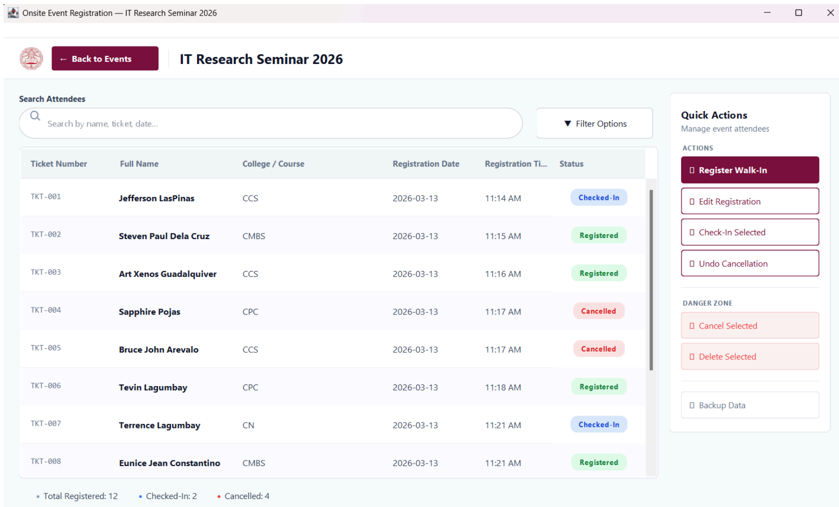

Create New Event
Create New Event
Fill in the details for your new event
Event Name *
Purpose / Description
Maximum Attendees (0 = Unlimited)
0
Enter 0 to allow unlimited registrations
Cancel
Create Event

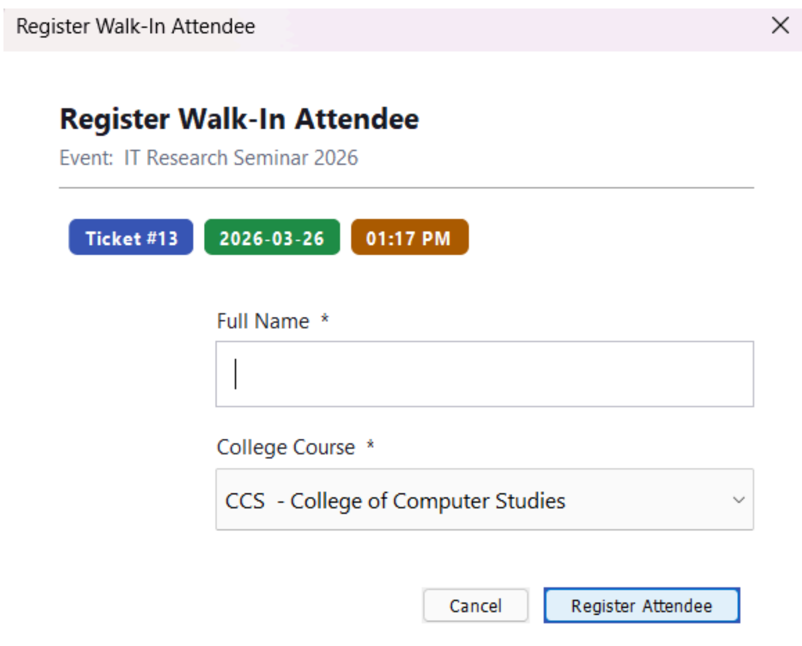
How to Edit an Existing Event:

1. On the HomeFrame (Home Dashboard), locate the event you wish to update in the events table.
2. Select the event and click the **"Edit Event"** button.
3. The **Edit Event Dialog** will appear on the screen.
4. You can now modify the following details:
   - **Event Name:** (e.g., change from "IT Seminar" to "CCS IT Seminar")
   - **Purpose:** Update the description or goal of the event.
   - **Date Created:** Ensure it strictly follows the YYYY-MM-DD format.
   - **Maximum Attendees:** Adjust the seating capacity. (Note: Must be a non-negative number; use '0' for unlimited).
5. Click the **"Save"** button. The system will validate your inputs and immediately update the event in the main dashboard.
6. 

3. Event Dashboard (Registering Attendees)

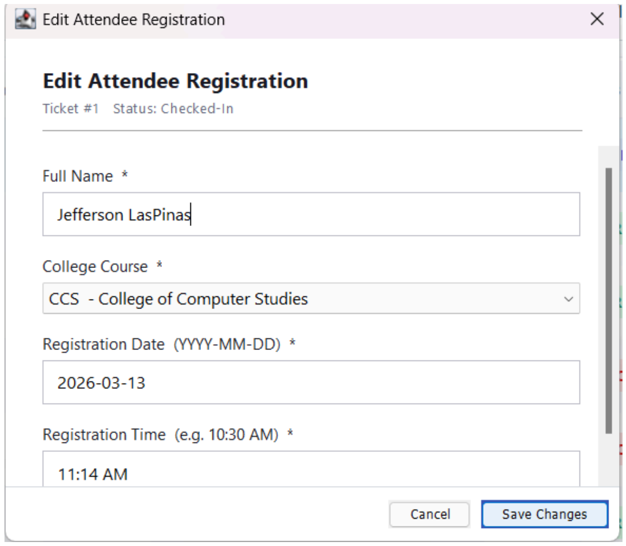
1. Inside the EventDashboardFrame, you will see the list of currently registered students and the remaining seating capacity.
2. **To Register a Walk-in:** Click **"Register Attendee"**. A form (RegisterAttendeeDialog) will pop up.
3. Enter the student's Full Name and select their College Course (e.g., CCS, CMBS) from the dropdown menu. The system will automatically log the time and date. Click "Register".

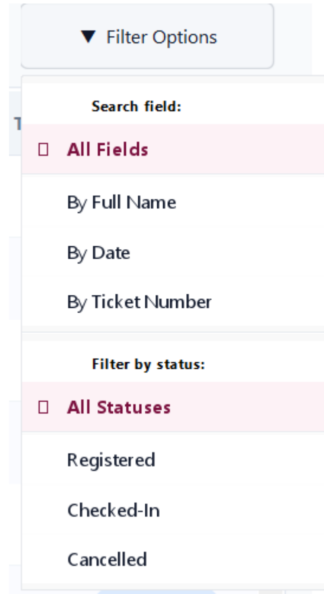
4. 



4. Managing Attendees (Edit & Cancel)

1. **To Edit:** If a student's name is misspelled or their course is wrong, select their row and click the **"Edit"** button to open the EditAttendeeDialog.
2. **To Cancel:** If a student leaves or wishes to cancel, select their row and click **"Cancel Attendee"**. This will update their status and free up one seat in the event capacity.
3. **To Filter:** Use the Status Dropdown above the table to view only "Registered", "Checked In", or "Cancelled" students.





5. Backup & Restore Feature The system provides a built-in Backup & Restore mechanism to protect your event data and recover attendee records if needed.

How to Create a Backup:

- Inside the Event Dashboard, open the "Actions" menu.
- Select the backup and restore option to open the dialog, then choose "Create Backup".
- The system will automatically take a snapshot of the current event data and save it into a timestamped folder within the **data/backups/** directory.

How to Restore a Backup:

- Open the "Actions" menu and access the backup and restore dialog, then select "Restore Backup".
- A new dialog will appear where you must first select the **Source event** you want to recover data from.
- Next, select the specific timestamped backup file you wish to load.
- Choose the attendees you want to recover from the list and proceed with the restoration.
- **Important Restoration Rules:** * **Capacity Limits:** The system validates all restored data to prevent overbooking; if the live event has reached its Maximum Attendees limit, excess attendees from the backup will be blocked from restoring.
    - **Duplicate Prevention:** The system actively checks for duplicates and will skip any names in the backup that are already actively registered in the current event.

- ○ **Status Reset:** Attendees who had a "Cancelled" status in the old backup will be re-admitted into the event as "Registered" by default.

C. Feature List

Our application provides complete CRUD (Create, Read, Update, Delete) functionality, alongside advanced filtering features:

- **Add (Create):** Operators can create new school events with strict maximum seating capacities. They can also add/register new walk-in students to these events, automatically generating a ticket number and timestamp.
- **View (Read):** The system displays events and attendees in responsive, scrollable Data Tables (JTable). It provides real-time statistics, such as total registered students versus remaining capacity.
- **Edit (Update):** Operators can update event details (like changing the venue capacity) and modify attendee records to fix typos or change selected college courses.
  Edit Event: This feature allows the system administrator or operator to modify the details of an already created event. It is particularly useful for dynamically adjusting the **Maximum Attendees** capacity if a venue changes, or for correcting typos in the event's name and date without needing to delete and recreate the entire event.
- **Delete / Cancel:** Operators can cancel a student's registration (which safely updates their status and recalculates available seating) or permanently delete a record/event if created by mistake.
- **Search & Filter:** A real-time composite search bar instantly filters records as the user types (searching across names, ticket numbers, and dates). A dropdown filter also allows sorting attendees by their active status.

D. Code Documentation

The application is built using Java Swing following Object-Oriented Programming (OOP) principles. Below are the key components of the codebase:

1. Important Classes

- **MainApp:** The entry point of the application. It safely launches the GUI on the Event Dispatch Thread using SwingUtilities.invokeLater().
- **LoginFrame, HomeFrame, EventDashboardFrame:** The primary GUI windows (JFrames) that handle user interaction, routing, and displaying the data tables.
- **Attendee & Event:** Model classes representing the core data structures. They contain encapsulated fields (variables), constructors, and Getter/Setter methods.
- **JsonHandler:** A utility class responsible for data persistence. It uses the org.json.simple library to save and load Java objects into local .json files, ensuring no data is lost when the app closes.

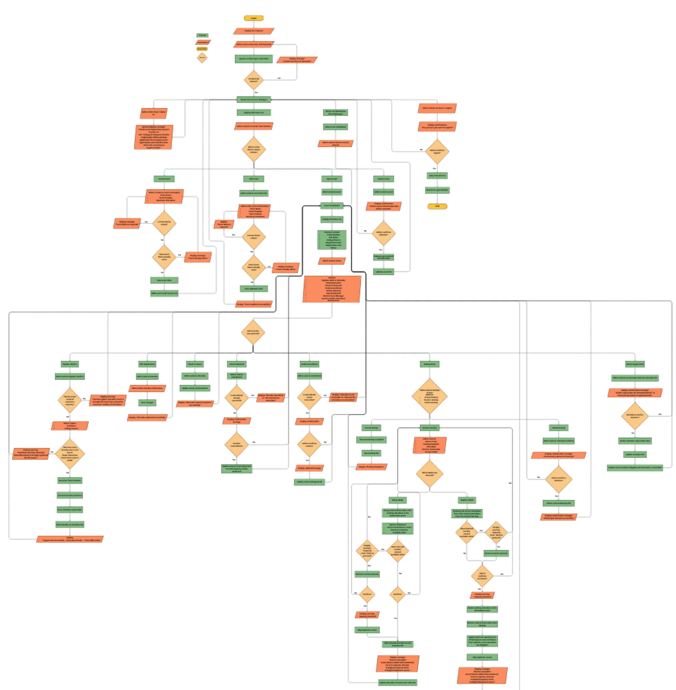
- **AttendeeValidationService:** The core business logic layer. It acts as the single source of truth for enforcing rules.

2. Key Methods & Logic

- **Validation Logic (AttendeeValidationService.validateMerge()):** This method contains the logic that prevents overbooking and duplicates. Before saving an attendee, it checks isCapacityFull() to ensure the event hasn't reached its maximum limit, and isDuplicate() to prevent registering the exact same student name twice.
- **Real-time Search Logic (DocumentListener):** Instead of a standard KeyListener or requiring a "Search" button, the text fields utilize a DocumentListener. This listens for any character changes in the search bar and triggers an applyFilterAndSort() method to instantly update the TableRowSorter of the JTable.
- **Method Overloading (Attendee.java):** The Attendee class utilizes method overloading with the getStatusLabel() method. It has multiple versions (one with no parameters, one with a boolean parameter, etc.) to format the attendee's display string differently depending on where it is shown in the UI.
- **Data Parsing (JsonHandler.loadEvents() / saveEvents()):**
  Uses try-catch blocks and FileReader/FileWriter to parse JSON arrays into Java ArrayLists.

You may access to this link for a full preview of this Flowchart:

https://www.canva.com/design/DAHEvZ3RaJA/IGNPgFlHFKaZvc6OjAGePA/edit



E. System Requirements

To ensure the system runs smoothly, the deployment environment must meet the following minimum requirements:

Software Requirements:

- **Java Version:** Java Development Kit (JDK) 8, 11, or 17 (Java 8 minimum is required to run Swing components and standard Time libraries).
- **Operating System:** Cross-platform (Windows 10/11, macOS, or Linux).
- **Dependencies:** `json-simple-1.1.1.jar` (must be included in the project classpath for data storage functionality).

Hardware Requirements:

- **Processor:** 1.0 GHz processor or faster (Intel/AMD).
- **RAM:** Minimum 512 MB (1 GB recommended).
- **Storage:** At least 50 MB of free disk space (to accommodate the application and the generated JSON data files).
- **Display:** Minimum resolution of 1024 x 768 for optimal UI scaling.

V. Comparative Analysis

Overview When organizing a school event, student councils and event staff must choose the right tool to manage attendees. Below is a comparison of our custom-built application against the two most common alternative solutions.

1. The Enterprise Application

- **App Name:** Eventbrite
- **Features:** Sells online tickets, scans QR codes, processes payments, and tracks marketing analytics.
- **Differences:** This is a heavy, commercial platform that requires a strong, constant internet connection. Because it is built for large-scale, paid events (like concerts or public conferences), setting it up for a simple, free school seminar takes too much time and unnecessary effort.

2. The Generic Cloud Form

- **App Name:** Google Forms (paired with Google Sheets)
- **Features:** Creates free online surveys and automatically saves the submitted answers to a cloud spreadsheet.
- **Differences:** This tool completely relies on the school's Wi-Fi. If the Wi-Fi is slow or goes down in a crowded gymnasium, you cannot register anyone. Additionally, Google Forms

will not automatically stop accepting answers when your venue is full, and it allows the same student to accidentally register twice.

3. Our Custom Solution

- **App Name:** School-Based Onsite Event Registration System
- **Features:** Fast walk-in registration, instant search bar, automatic seat limits, duplicate name blocking, and offline data saving.
- **Differences:** Unlike the other tools, this is a **100% offline desktop app**. It runs directly on the registration desk's laptop, meaning it works perfectly even if the school internet goes down.
- **What Makes Our System Unique:** It is built specifically for the fast-paced chaos of school events. Because it works offline, the registration line never slows down. Furthermore, it automatically solves the two biggest headaches for student organizers: it instantly prevents a student from registering twice, and it automatically locks down registrations the exact second the room reaches its maximum seating capacity.

VI. Testing and Debugging

TEST CASE	PRE-CONDITIO N(S)	PROCEDURE(S)	SAMPLE TEST DATA	EXPECTED RESULT	ACTUAL RESULT	STATUS
Empty login fields	Application must be running; Login JFrame visible.	Click Login without input	No username and password input	The system denied access to the main Dashboard.	Invalid username and password	Passed
Invalid credentials	Application must be running; Login JFrame visible.	1. Enter different Username 2. Enter different Password 3. Click Login	User: jeff Pass:jeff123	The system denied access to the main Dashboard.	Invalid username and password	Passed
Admin Login	Application must be running; Login JFrame visible.	1. Enter Username 2. Enter Password 3. Click Login	User: admin Pass:admin123	The system grants access to the main Dashboard.	Access granted	Passed
Add Event	Dashboard is visible; Event function is	1. Enter Event Name	Name: IT Research Seminar 2026	Event details appear in the table; JSON file	Table updated with all fields	Passed

TEST CASE	PRE-CONDITIO N(S)	PROCEDURE(S)	SAMPLE TEST DATA	EXPECTED RESULT	ACTUAL RESULT	STATUS
	accessible.	2.Enter Event Purpose 3.Add maximum Attendees	Purpose: Annual presentation Max: 100 Date: 2026-03-23	updates with Name, Purpose, Max, and Date Created.		
Edit Event	An existing event must be selected from the table.	1. Click "Edit" on selected event 2. Modify Name, Purpose, Date, or Max 3. Click Save Changes	New Name: IT Expo 2026 New Purpose: Tech Showcase New Max: 150 New Date: 2026-04-1	The table reflects the new values and the JSON file is overwritten with updated info.	Record updated successfully	**Passed**
Delete Event	At least one event must exist in the list.	1. Select event from table 2. Click Delete button 3. Click "Yes" on Confirmation Dial	Selection: IT Expo 2026 Confirm: Yes	A pop-up asks **"Delete event "IT Expo 2026"?** All attendee data will be permanently removed confirmation, the record is removed from the table and JSON.	Record removed after confirmation	**Passed**
Event Name Duplication	Existing record	Enter same Event Name	Name: IT Research Seminar 2026	Event details appear in the table; JSON file updates the Event Name.	Warning An event name duplication detected Name: IT Research Seminar 2026 Is already created	**Passed**

TEST CASE	PRE-CONDITION(S)	PROCEDURE(S)	SAMPLE TEST DATA	EXPECTED RESULT	ACTUAL RESULT	STATUS
Edit Event Name Duplication	Existing record	Enter same Event Name	Name: IT Research Seminar 2026	Event details appear in the table; JSON file updates with Event Name..	Warning An event name duplication detected Name: IT Research Seminar 2026 Is already exists	Passed
Search bar (Event)	The search bar is visible; multiple events exist in JSON.	1. Type event name in search bar 2. System filters automatically or on Enter	**Search:** IT Research	The table dynamically filters to show only the matching event; JSON remains unchanged.	Table filtered	Passed
Sort by	Multiple events with different dates must exist.	1. Click Sort dropdown 2. Select **"Date: Newest to Oldest"**	**Sort:** Date (Newest)	Table reorders events chronologically based on the Date Created	Order changed	Passed
Register Walk-in	Registration module is active	. Enter Full Name 2. Select College Course 3. System auto-generates Date/Time 4. Click Register	Name: Bruce John B. Arevalo Course: CCS Status: Registered	Attendee added to table with Reg Date, Reg Time, and Status. JSON file updates	Record added with timestamp	Passed

TEST CASE	PRE-CONDITIO N(S)	PROCEDURE(S)	SAMPLE TEST DATA	EXPECTED RESULT	ACTUAL RESULT	STATUS
Register Duplicate attendee	Existing record	Enter same Full Name	Name: Bruce John B. Arevalo Course: CCS Status: Registered	Attendee added to table with Reg Date, Reg Time, and Status. JSON file updates	Warning Duplication attendee detected Name: Bruce John B. Arevalo Is already registered	Passed
Event Registration in max attendees	Max reached	1. Enter Full Name 2. Select College Course 3. System auto-generates Date/Time 4. Click Register	Name: Bruce John B. Arevalo Course: CCS Status: Registered	Attendee added to table with Reg Date, Reg Time, and Status. JSON file updates	Warning this event has reached its maximum attendees	Passed
Edit Registration	Attendee records must exist in the system..	1. Select Attendee 2. Modify Full Name, Course, Reg Date, or Reg Time 3. Click Update	Edited Name New Date: 2026-03-24 New Time: 09:30 AM New Course: CPC	The table updates and the JSON file reflects the changed details across all four fields.	Info updated successfully	Passed
Edit name Duplicate attendee	Attendee records must exist in the system..	1. Select Attendee 2. Modify Full Name 3. Click Update	Edited name	The table updates and the JSON file reflects the changed details across all four fields.	Warning Duplication attendee detected	Passed

TEST CASE	PRE-CONDITION(S)	PROCEDURE(S)	SAMPLE TEST DATA	EXPECTED RESULT	ACTUAL RESULT	STATUS
Checked-In Selected	One or more rows must be selected in the table.	1. Highlight specific row(s) 2. Click Select Checked-In	Selection: Bruce John Arevalo	Attendee registration status will turn to checked-in status	Attendee's status has been checked-in	Passed
Cancel Selected	One or more rows must be selected in the table.	1. Highlight specific row(s) 2. Click Select Cancel	Selection: Bruce John Arevalo	Attendee registration status will turn to cancel status	Attendee's status has been cancelled	Passed
Undo Cancellation	A row must be currently on cancel status	1. Click Undo Cancellation	Action: Undoing cancellation	Let the attendee to undo his cancellation status	Attendee status change back to registered	Passed
Delete Selected	One or more rows must be selected in the table.	1. Highlight specific row(s) 2. Click Select Delete	Selection: Bruce John Arevalo	Attendee's data will be permanently deleted	Attendee's data has been deleted	Passed
Create Backup	An event exists in the system; "Backup & Restore" prompt is active.	1. Select the target event. 2. Click the "Create Backup" button. 3. Confirm the backup creation.	Event: IT Research Seminar 2026	Seminar 2026: A new backup snapshot is created; confirmation message appears.	Backup snapshot generated successfully	Passed
Delete Backups	The system contains multiple existing backup snapshots (e.g., as seen in the list below). The user has access to the backup management screen.	Click "Delete Backups". From the list, select multiple backup snapshots using Ctrl+Click (e.g., "IT Research Seminar 2026" and "IOT	Available Backups for Deletion: IT Research Seminar 2026 — 2026-03-23 10:51:40 (13 attendees)	The selected backup snapshots are permanently deleted and cannot be recovered. The remaining backups (e.g., "IT Research Seminar 2026") remain listed. A confirmation	The selected backup snapshots (**"IT Research Seminar 2026"** were permanently deleted and are no longer visible in the list and on the JSON File.	Passed

TEST CASE	PRE-CONDITIO N(S)	PROCEDURE(S)	SAMPLE TEST DATA	EXPECTED RESULT	ACTUAL RESULT	STATUS
		Seminar 2026"). Click "OK" to confirm deletion.		message confirms the deletion.		
Restore Data	At least one backup snapshot must exist for the source event.	1. Click "Restore from Backup". 2. Select source event and snapshot. 3. Select attendees to restore. 4. Select target "Restore INTO" event. 5. Click "Restore Settings" (or Confirm).	Source: IOT Seminar 2 Target: IT Research Seminar 2026 Mode: Merge	Selected attendees are successfully added/merged into the target event list.	Attendees restored and merged into the table	Passed
Restore Data	At least one backup snapshot must exist for the source event.	1. Click "Restore from Backup". 2. Select source event and snapshot. 3. Select attendees to restore. 4. Select target "Restore INTO" event. 5. Click "Restore Settings" (or Confirm).	Source: IOT Seminar 2 Target: IT Research Seminar 2026 Mode: Replace	All existing attendees in the **"IT Research Seminar 2026"** list are removed. The list is successfully **overwritten** with the attendees from the **"IOT Seminar 2"** backup snapshot.	The current event list was cleared and replaced by the backup data. Only the attendees from the **"IOT Seminar 2"** snapshot are now visible in the table.	Passed
Restore exceeding capacity	Event Full	Restore backup	Restore data of Bruce John Arevalo	Only allowed slots restored	Warning Capacity reached its max capacity	Passed

TEST CASE	PRE-CONDITIO N(S)	PROCEDURE(S)	SAMPLE TEST DATA	EXPECTED RESULT	ACTUAL RESULT	STATUS
Filter Options	Verify that the Status Dropdown correctly filters the attendee table based on the selected status.	1. The user is logged into the application. 2. The user has navigated to an **Event Dashboard**. 3. The event has multiple attendees with different statuses ("Registered", "Checked In", and "Cancelled").	1. Locate the Status filter dropdown menu above the attendee table. 2. Click the dropdown and select **"Registered"**. 3. Click the dropdown and select **"Checked In"**. 4. Click the dropdown and select **"Cancelled"**. 5. Click the dropdown and select **"All Statuses"**.	- Selecting a specific status should immediately hide all non-matching attendees and display only those with the chosen status. - Selecting "All Statuses" should reset the filter and display the complete list of attendees.	The table correctly updated instantly upon each selection. "Registered", "Checked In", and "Cancelled" successfully isolated their respective records. "All Statuses" restored the full attendee list.	Passed

VII. Reflection and Enhancements - INDIVIDUAL

Jefferson Laspiñas:

Role: Designed and built the **Login Frame**, **HomeFrame/EventManager, EventDashboard Frame**, and **Register Walk-In Form** using Java Swing.

Reflection: “Building this project helped me understand how Java Swing components, event handling, validation logic, and JSON file handling work together in a real application. I learned how important it is to separate the UI, logic, and data layers so the system is easier to manage and debug. I also experienced real challenges such as fixing UI bugs, handling restore capacity issues, and converting dialogs to JFrame without breaking functionality. These problems helped me improve my debugging skills and understand how different parts of the system are connected. Overall, the project gave me hands-on experience in designing a functional desktop application using Java Swing and applying proper architecture and validation techniques.”

Proposed Enhancement: For future versions of the system, several enhancements can be implemented to improve performance, usability, and reliability. One major improvement is replacing the current JSON file storage with a proper database such as MySQL or SQLite to handle larger amounts of attendee data more efficiently and securely. The system can also include user roles and accounts, allowing administrators and staff to have different access permissions. Additional features like exporting the attendee list to Excel or PDF can make reporting and documentation easier. A QR code–based check-in system can be introduced to speed up attendee verification during events. The user interface can be modernized by transitioning from Swing to JavaFX for a cleaner and more responsive design. Search, filter, and sorting functionalities in the dashboard can be enhanced for better data management. Finally, adding an auto-save and auto-backup feature will help prevent data loss and make the system more reliable in case of unexpected errors or crashes.

Karl Andrei De Mira:

Role: Project Implementation/Planning and Documentation. Coordinated the development lifecycle, ensured backend-UI alignment, and managed all technical documentation and presentation materials.

Reflection: implementing this project highlighted the "big picture" perspective. Coordinating modules like JSON handling and validation required deep oversight to ensure the system functioned as a unified tool. It strengthened my ability to translate technical code into clear communication.

Proposed Enhancement: Integration of an **Administrative Analytics Dashboard** to generate visual data insights (e.g., peak registration times) from the existing JSON data.

Gian Galagar:

Role: Flowchart Designer and Lead Debugger. Mapped system logic through flowcharts and conducted stress testing to identify edge cases like capacity overflows.

Reflection: Working on the flowchart and testing the system flow help me better understand how each part of the program is connected. By analyzing the flow of the application, I was able to spot issues in navigation and logic that could affect the user experience. This role improved my attention to detail and strengthened my understanding of how proper planning and testing are important in developing a functional and reliable system.

Proposed Enhancement: QR Code Integration using the ZXing library to scan student IDs for faster, error-free registration during high-volume events. I suggest integrating QR Code Integration. By adding a library like ZXing, the system could scan student IDs to auto-fill the registration form, which would drastically reduce human error during high-volume walk-ins.